Applied Calculus IV

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Week 14

Chapter 7

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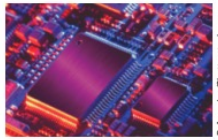

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Definition (Laplace Transform)

Let f be a function defined for $t \geq 0$. The **Laplace transform** of f, denoted $\mathcal{L}\{f(x)\}$, is given by:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\, dt$$

when the integral **converges**.

Exercise: Show that if $\mathscr{L}\{f(t)\} = F(s)$ and a is any real number, then

$$\mathscr{L}\{e^{at} f(t)\} = F(s-a).$$

Solution:

By definition,

$$\mathscr{L}\{e^{at}f(t)\} = \int_0^\infty e^{-st}e^{at}f(t)\,dt = \int_0^\infty e^{-(s-a)t}f(t)\,dt = F(s-a).$$

Example: Evaluate $\mathcal{L}\{e^{5t}t^3\}$

Solution:

$$\mathcal{L}\{t^3\} = \frac{3!}{s^4} \xrightarrow{s \to s-5} \mathcal{L}\{e^{5t}t^3\} = \frac{3!}{(s-5)^4}$$

Example: Evaluate $\mathcal{L}\{e^{-2t}\cos 4t\}$

Solution:

$$\mathcal{L}\{\cos 4t\} = \frac{s}{s^2+16} \xrightarrow{s \to s+2} \mathcal{L}\{e^{-2t}\cos 4t\} = \frac{(s+2)}{(s+2)^2+16}$$

Example: Evaluate

$$\mathcal{L}^{-1}\left\{\frac{7s+4}{s^2+4s+6}\right\}$$

Solution:

$$\mathcal{L}^{-1}\left\{\frac{7s+4}{s^2+4s+6}\right\} = \mathcal{L}^{-1}\left\{\frac{7s+14-10}{(s+2)^2+2}\right\} = 7\mathcal{L}^{-1}\left\{\frac{s+2}{(s+2)^2+2}\right\} - 10\mathcal{L}^{-1}\left\{\frac{1}{(s+2)^2+2}\right\}$$

$s+2 \to s$

$s+2 \to s$

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2+2}\right\} = \cos\sqrt{2}t$$

$$\mathcal{L}^{-1}\left\{\frac{\sqrt{2}}{s^2+2}\right\} = \sin\sqrt{2}t$$

$$\Rightarrow \quad \mathcal{L}^{-1}\left\{\frac{s+5}{s^2+4s+6}\right\} = 7e^{-2t}\cos\sqrt{2}t - 5\sqrt{2}e^{-2t}\sin\sqrt{2}t$$

Example: Evaluate

$$\mathcal{L}^{-1}\left\{\frac{2s+5}{(s-3)^2}\right\}$$

Solution:

$$\mathcal{L}^{-1}\left\{\frac{2s+5}{(s-3)^2}\right\} = \mathcal{L}^{-1}\left\{\frac{2s-6+11}{(s-3)^2}\right\} = 2\mathcal{L}^{-1}\left\{\frac{\cancel{s-3}}{(s-3)^2}\right\} + 11\mathcal{L}^{-1}\left\{\frac{1}{(s-3)^2}\right\}$$

$$\frac{1}{s-3}$$

$$s-3 \to s \qquad\qquad s-3 \to s$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1 \qquad\qquad \mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = t$$

$$\Rightarrow \quad \mathcal{L}^{-1}\left\{\frac{2s+5}{(s-3)^2}\right\} = 2e^{3t} + 11te^{3t}$$

In Problem 1-10, evaluate $F(s)$

1. $\mathscr{L}\{te^{10t}\}$
2. $\mathscr{L}\{te^{-6t}\}$
3. $\mathscr{L}\{t^3e^{-2t}\}$
4. $\mathscr{L}\{t^{10}e^{-7t}\}$
5. $\mathscr{L}\{t(e^t + e^{2t})^2\}$
6. $\mathscr{L}\{e^{2t}(t - 1)^2\}$
7. $\mathscr{L}\{e^t \sin 3t\}$
8. $\mathscr{L}\{e^{-2t} \cos 4t\}$
9. $\mathscr{L}\{(1 - e^t + 3e^{-4t}) \cos 5t\}$
10. $\mathscr{L}\left\{e^{3t}\left(9 - 4t + 10 \sin \frac{t}{2}\right)\right\}$

In Problem 10-20, evaluate $f(t)$

11. $\mathscr{L}^{-1}\left\{\dfrac{1}{(s+2)^3}\right\}$

12. $\mathscr{L}^{-1}\left\{\dfrac{1}{(s-1)^4}\right\}$

13. $\mathscr{L}^{-1}\left\{\dfrac{1}{s^2-6s+10}\right\}$

14. $\mathscr{L}^{-1}\left\{\dfrac{1}{s^2+2s+5}\right\}$

15. $\mathscr{L}^{-1}\left\{\dfrac{s}{s^2+4s+5}\right\}$

16. $\mathscr{L}^{-1}\left\{\dfrac{2s+5}{s^2+6s+34}\right\}$

17. $\mathscr{L}^{-1}\left\{\dfrac{s}{(s+1)^2}\right\}$

18. $\mathscr{L}^{-1}\left\{\dfrac{5s}{(s-2)^2}\right\}$

19. $\mathscr{L}^{-1}\left\{\dfrac{2s-1}{s^2(s+1)^3}\right\}$

20. $\mathscr{L}^{-1}\left\{\dfrac{(s+1)^2}{(s+2)^4}\right\}$

More Examples …

Example: Solve the equation $x'' + 2x' + x = e^{-t}$, with $x(1) = x'(1) = 0$.

Recall

$$\mathcal{L}[x'] = s\mathcal{L}[x] - x(0)$$

$$\mathcal{L}[x''] = s^2\mathcal{L}[x] - sx(0) - x'(0)$$

Solution:

Let $\tau = t - 1$:

$$x''(\tau) + 2x'(\tau) + x(\tau) = e^{-(\tau+1)}, \quad x(0) = x'(0) = 0.$$

Take the Laplace transform:

$$s^2X(s) + 2sX(s) + X(s) = \frac{1}{s+1} \cdot \frac{1}{e} \quad \Longrightarrow \quad X(s) = \boxed{\frac{1}{(s+1)^3}} \cdot \frac{1}{e}$$

$s + 1 \to s$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\} = \frac{1}{2}\tau^2 \longrightarrow \mathcal{L}^{-1}\left\{\frac{1}{(s+1)^3}\right\} = \frac{1}{2}e^{-t}\tau^2$$

$$x(\tau) = \frac{1}{2}\tau^2 e^{-\tau-1}$$

Solution:

And finally, $x(t)$ is

$$x(t) = \frac{1}{2}(t-1)^2 e^{-t}.$$

Example: Use the Laplace transform to solve the initial-value problem

$$x''' + 3x'' + 3x' + x = 1, \quad x(0) = 0,\ x'(0) = 0,\ x''(0) = 0$$

Recall

$$\mathcal{L}\{x'(t)\} = sX(s) - x(0)$$

$$\mathcal{L}\{x''(t)\} = s^2X(s) - sx(0) - x'(0)$$

$$\mathcal{L}\{x'''(t)\} = s^3X(s) - s^2x(0) - sx'(0) - x''(0)$$

Solution:

Since $x(0) = x'(0) = x''(0) = 0,$

$$x''' + 3x'' + 3x' + x = 1$$

$$\xrightarrow{\mathcal{L}}$$

$$\underbrace{[s^3 X(s)]}_{x'''} + 3\underbrace{[s^2 X(s)]}_{x''} + 3\underbrace{[sX(s)]}_{x'} + \underbrace{X(s)}_{x} = \frac{1}{s}$$

$$X(s)\left(s^3 + 3s^2 + 3s + 1\right) = \frac{1}{s}$$

$$\Rightarrow \quad X(s) = \frac{1}{s(s+1)^3}$$

Solution:

Decompose the fraction $$\frac{1}{s(s+1)^3} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{(s+1)^2} + \frac{D}{(s+1)^3}$$

$$X(s) = \frac{1}{s} - \frac{1}{s+1} - \frac{1}{(s+1)^2} - \frac{1}{(s+1)^3}$$

$\mathcal{L}^{-1}$

$$x(t) = 1 - e^{-t} - te^{-t} - \frac{1}{2}t^2e^{-t}$$

Example: Use Laplace transforms to solve the given system of differential equations

$$\begin{cases} x_1' = 3x_1 + 5x_2 + e^{-t}, \\ x_2' = -5x_1 + 3x_2 \end{cases}$$

with the initial conditions

$$x_1(0) = 0,\ x_2(0) = 1$$

Solution:

Let $X_1(s) = \mathcal{L}[x_1(t)]$ and $X_2(s) = \mathcal{L}[x_2(t)]$

$$\begin{cases} x_1' = 3x_1 + 5x_2 + e^{-t}, \\ x_2' = -5x_1 + 3x_2 \end{cases} \Rightarrow \begin{cases} sX_1(s) = 3X_1(s) + 5X_2(s) + \dfrac{1}{s+1}, \\ sX_2(s) - 1 = -5X_1(s) + 3X_2(s) \end{cases}$$

or

$$\begin{cases} (s-3)X_1(s) - 5X_2(s) = \dfrac{1}{s+1}, \\ 5X_1(s) + (s-3)X_2(s) = 1 \end{cases}$$

Cramer's Rule ($A\mathbf{X} = \mathbf{b}$)

$$A = \begin{pmatrix} s-3 & -5 \\ 5 & s-3 \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} \dfrac{1}{s+1} \\ 1 \end{pmatrix}$$

$$A_1 = \begin{pmatrix} \dfrac{1}{s+1} & -5 \\ 1 & s-3 \end{pmatrix} \Rightarrow X_1 = \frac{\det(A_1)}{\det(A)} = \frac{\dfrac{s-3}{s+1} + 5}{(s-3)^2 + 5^2},$$

$$A_2 = \begin{pmatrix} s-3 & \dfrac{1}{s+1} \\ 5 & 1 \end{pmatrix} \Rightarrow X_2 = \frac{\det(A_2)}{\det(A)} = \frac{s-3-\dfrac{5}{s+1}}{(s-3)^2 + 5^2}$$

$$\Rightarrow \begin{cases} X_1(s) = \dfrac{\dfrac{s-3}{s+1} + 5}{(s-3)^2 + 5^2} \\ X_2(s) = \dfrac{s-3-\dfrac{5}{s+1}}{(s-3)^2 + 5^2} \end{cases}$$

$$\begin{cases} X_1(s) = \dfrac{6s+2}{(s+1)\left[(s-3)^2+25\right]} \\[2ex] X_2(s) = \dfrac{(s-4)(s+2)}{(s+1)\left[(s-3)^2+25\right]} \end{cases}$$

Decompose the fractions

$$\begin{cases} X_1(s) = \dfrac{1}{41}\left[4\dfrac{s-3}{(s-3)^2+5^2} + 46\dfrac{5}{(s-3)^2+5^2} - 4\dfrac{1}{s+1}\right] \\[2ex] X_2(s) = \dfrac{1}{41}\left[46\dfrac{s-3}{(s-3)^2+5^2} - 4\dfrac{5}{(s-3)^2+5^2} - 5\dfrac{1}{s+1}\right] \end{cases}$$

$\mathcal{L}^{-1}$

$$\begin{cases} x_1(t) = \dfrac{4}{41}e^{3t}\cos 5t + \dfrac{46}{41}e^{3t}\sin 5t - \dfrac{4}{41}e^{-t}, \\[2ex] x_2(t) = \dfrac{46}{41}e^{3t}\cos 5t - \dfrac{4}{41}e^{3t}\sin 5t - \dfrac{5}{41}e^{-t} \end{cases}$$

More details:

$$X_1(s) = \frac{1}{41}\left[4\frac{s-3}{(s-3)^2+5^2} + 46\frac{5}{(s-3)^2+5^2} - 4\frac{1}{s+1}\right]$$

$s-3 \rightarrow s$

$s-3 \rightarrow s$

$$\mathcal{L}^{-1}\left\{\frac{1}{s-(-1)}\right\} = e^{-t}$$

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2+5^2}\right\} = \cos 5t$$

$$\mathcal{L}^{-1}\left\{\frac{5}{s^2+5^2}\right\} = \sin 5t$$

$$\mathcal{L}^{-1}\left\{\frac{s-3}{(s-3)^2+5^2}\right\} = e^{3t}\cos 5t$$

$$\mathcal{L}^{-1}\left\{\frac{5}{(s-3)^2+5^2}\right\} = e^{3t}\sin 5t$$

$$x_1(t) = \frac{4}{41}e^{3t}\cos 5t + \frac{46}{41}e^{3t}\sin 5t - \frac{4}{41}e^{-t}$$

And similarly for $X_2(s)$

Example: Use Laplace transforms to solve the given system of differential equations

$$\begin{cases} x_1' = 2x_1 + x_2 \\ x_2' = -x_1 + 4x_2 \end{cases}$$

with the initial conditions

$$x_1(0) = 0,\ x_2(0) = 1$$

Solution:

Let $X_1(s) = \mathcal{L}[x_1(t)]$ and $X_2(s) = \mathcal{L}[x_2(t)]$

$$\begin{cases} x_1' = 2x_1 + x_2 \\ x_2' = -x_1 + 4x_2 \end{cases}$$

$\mathcal{L}$

$$\begin{cases} sX_1(s) - \cancel{x_1(0)} = 2X_1(s) + X_2(s) \\ sX_2(s) - \underset{1}{\cancel{x_2(0)}} = -X_1(s) + 4X_2(s) \end{cases} \Rightarrow \begin{cases} (s-2)X_1(s) - X_2(s) = 0 \\ X_1(s) + (s-4)X_2(s) = 1 \end{cases}$$

Cramer's Rule ($A\mathbf{X} = \mathbf{b}$)

$$\det(A) = \begin{vmatrix} s-2 & -1 \\ 1 & s-4 \end{vmatrix} = (s-2)(s-4)+1 = s^2-6s+9 = (s-3)^2$$

$$X_1(s) = \frac{\begin{vmatrix} 0 & -1 \\ 1 & s-4 \end{vmatrix}}{\det(A)} = \frac{0\cdot(s-4)-(-1)\cdot 1}{(s-3)^2} = \frac{1}{(s-3)^2}$$

$$X_2(s) = \frac{\begin{vmatrix} s-2 & 0 \\ 1 & 1 \end{vmatrix}}{\det(A)} = \frac{(s-2)\cdot 1-0\cdot 1}{(s-3)^2} = \cancel{\frac{s-2}{(s-3)^2}}$$

$$\frac{(s-3)+1}{(s-3)^2} = \frac{1}{s-3} + \frac{1}{(s-3)^2}$$

$$x_1(t) = \mathcal{L}^{-1}\left\{\frac{1}{(s-3)^2}\right\} = te^{3t}$$

$$x_2(t) = \mathcal{L}^{-1}\left\{\frac{1}{s-3} + \frac{1}{(s-3)^2}\right\} = e^{3t} + te^{3t} = e^{3t}(1+t)$$

$$\Rightarrow \begin{cases} x_1(t) = te^{3t} \\ x_2(t) = e^{3t}(1+t) \end{cases}$$

is the solution of the given IVP

Example: Use Laplace transforms to solve the given system of differential equations

$$\begin{cases} x_1'' - 2x_1' - x_2' + 2x_2 = 0 \\ x_1' - 2x_1 + x_2' = -2e^{-t} \end{cases}$$

with the initial conditions

$$x_1(0) = 3, \quad x_1'(0) = 2, \quad x_2(0) = 0$$

Solution:

$$\mathcal{L}\{x_1''\} = s^2X_1(s) - sx_1(0) - x_1'(0)$$

$$\mathcal{L}\{x_1'\} = sX_1(s) - x_1(0)$$

$$\mathcal{L}\{x_2'\} = sX_2(s) - x_2(0)$$

$$\mathcal{L}\{x_2\} = X_2(s)$$

Replacement in $\begin{cases} x_1'' - 2x_1' - x_2' + 2x_2 = 0 \\ x_1' - 2x_1 + x_2' = -2e^{-t} \end{cases}$

$$\left[s^2X_1(s) - 3s - 2\right] - 2\left[sX_1(s) - 3\right] - \left[sX_2(s) - 0\right] + 2X_2(s) = 0$$

$$\left[sX_1(s) - 3\right] - 2X_1(s) + \left[sX_2(s) - 0\right] = -\tfrac{2}{s+1}$$

$$\begin{cases} (s^2 - 2s)X_1(s) - (s-2)X_2(s) = 3s - 4 \\ (s-2)X_1(s) + sX_2(s) = \dfrac{3s+1}{s+1} \end{cases}$$

$$\times s \quad \begin{cases} (s^2 - 2s)X_1(s) - (s-2)X_2(s) = 3s - 4 \\ (s-2)X_1(s) + sX_2(s) = \dfrac{3s+1}{s+1} \end{cases}$$

$$-X_2(s^2 + s - 2) = \frac{-2s-4}{s+1} \longrightarrow X_2(s+2)(s-1) = \frac{2(s+2)}{s+1}$$

$$X_2(s) = \frac{2}{(s-1)(s+1)} \qquad \text{for } s \neq -2$$

or $$X_2(s) = \frac{2}{s^2 - 1}$$

- You also can use Cramer's Rule $\longrightarrow$ $X_1(s) = \dfrac{3s^2 - 4s - 1}{(s-2)(s^2-1)}$

Decompose the fractions

$$\frac{3s^2-4s-1}{(s-2)(s-1)(s+1)}=\frac{A}{s-2}+\frac{B}{s-1}+\frac{C}{s+1}$$

$$\frac{2}{(s-1)(s+1)}=\frac{A}{s-1}+\frac{B}{s+1}$$

$$X_1(s)=\frac{1}{s-2}+\frac{1}{s-1}+\frac{1}{s+1} \longrightarrow x_1(t)=e^{2t}+e^{t}+e^{-t}$$

$$X_2(s)=\frac{1}{s-1}-\frac{1}{s+1} \longrightarrow x_2(t)=e^{t}-e^{-t}$$

$$\Rightarrow \begin{cases} x_1(t)=e^{t}+e^{2t}+e^{-t} \\ x_2(t)=e^{t}-e^{-t} \end{cases}$$

In-Class Exercises

Exercise 1: Use the Laplace transform to solve the initial-value problem

$$y' + y = e^{-3t} \cos 2t, \quad y(0) = 0$$

Recall

$$\boxed{\mathcal{L}\{y'(t)\} = sY(s) - y(0)}$$

Solution:

$$y' + y = e^{-3t}\cos 2t,$$

$\mathcal{L}$

$$sY(s) + Y(s) = \frac{s+3}{(s+3)^2+4} \longrightarrow Y(s)(s+1) = \frac{s+3}{(s+3)^2+4}$$

$$\longrightarrow Y(s) = \frac{s+3}{(s+1)\left[(s+3)^2+4\right]}$$

Decompose it

$$\frac{A}{s+1} + \frac{B(s+3)+C}{(s+3)^2+4}$$

$$Y(s)=\frac{1/4}{s+1}+\frac{-\frac{1}{4}(s+3)+\frac{1}{2}}{(s+3)^2+4}$$

$$Y(s)=\frac{1}{4}\cdot\frac{1}{s+1}-\frac{1}{4}\cdot\frac{s+3}{(s+3)^2+4}+\frac{1}{2}\cdot\frac{1}{(s+3)^2+4} \xrightarrow{\mathcal{L}^{-1}} ?$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s-(-1)}\right\}=e^{-t}$$

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2+2^2}\right\}=\cos 2t$$

$$\mathcal{L}^{-1}\left\{\frac{2}{s^2+2^2}\right\}=\sin 2t$$

Finally,

$$\boxed{y(t) = \frac{1}{4}e^{-t} + \frac{1}{4}e^{-3t}\left(\sin 2t - \cos 2t\right)}$$ The solution of the given IVP

Exercise 2: Use the Laplace transform to solve the initial-value problem

$$y'' + 2y = 4t, \quad y(0) = 0, \quad y'(0) = 0$$

Recall

$$\mathcal{L}\{y'(t)\} = sY(s) - y(0)$$

$$\mathcal{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0)$$

Solution:

$$y'' + 2y = 4t,$$

$$\mathcal{L}$$

$$s^2Y(s) + 2Y(s) = \frac{4}{s^2} \implies Y(s) = \frac{4}{s^2(s^2+2)} \xrightarrow{\text{Decompose it}} \frac{As+B}{s^2} + \frac{Cs+D}{s^2+2}$$

$$Y(s) = \frac{2}{s^2} - \frac{2}{s^2+2}$$

$$\mathcal{L}^{-1}$$

$$y(t) = 2t - \sqrt{2}\sin(\sqrt{2}\,t)$$

Exercise 3: Use the Laplace transform to solve the initial-value problem

$$y'' + 4y' + 6y = 1 + e^{-t}, \quad y(0) = 0, \quad y'(0) = 0.$$

Recall

$$\mathcal{L}\{y'(t)\} = sY(s) - y(0)$$

$$\mathcal{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0)$$

Solution:

$$y'' + 4y' + 6y = 1 + e^{-t},$$

$\mathcal{L}$

$$s^2Y(s) - s\cancel{y}(0) - \cancel{y'}(0) + 4[sY(s) - \cancel{y}(0)] + 6Y(s) = \frac{1}{s} + \frac{1}{s+1}$$

Simplify it

$$(s^2 + 4s + 6)Y(s) = \frac{2s+1}{s(s+1)}$$

$$Y(s) = \frac{2s+1}{s(s+1)(s^2+4s+6)}$$

Decompose it

$$\frac{A}{s} + \frac{B}{s+1} + \frac{C(s+2)+D}{(s+2)^2+2}$$

$$Y(s)=\frac{1/6}{s}+\frac{1/3}{s+1}+\frac{-\frac{1}{2}(s+2)-\frac{2}{3}}{(s+2)^2+2}$$

$\mathcal{L}^{-1}$

$$y(t)=\frac{1}{6}\mathcal{L}^{-1}\left\{\frac{1}{s}\right\}+\frac{1}{3}\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\}-\frac{1}{2}\mathcal{L}^{-1}\left\{\frac{s+2}{(s+2)^2+2}\right\}-\frac{2}{3\sqrt{2}}\mathcal{L}^{-1}\left\{\frac{\sqrt{2}}{(s+2)^2+2}\right\}$$

$$=\frac{1}{6}+\frac{1}{3}e^{-t}-\frac{1}{2}e^{-2t}\cos\sqrt{2}\,t-\frac{\sqrt{2}}{3}e^{-2t}\sin\sqrt{2}\,t.$$

More Exercises …

Exercise 1: Find the solution to the following initial value problems

(1) $x'' + 9x = 6e^{3t}, \quad x(0) = x'(0) = 0;$

(2) $x^{(4)} + x = 2e^{t}, \quad x(0) = x'(0) = x''(0) = x'''(0) = 1.$

Exercise 2: Find the solution to the following initial value problems

$$(1)\ \begin{cases} x_1' + x_2' = 0, \\ x_1' - x_2' = 1, \end{cases} \qquad x_1(0) = 1,\ x_2(0) = 0;$$

$$(2)\ \begin{cases} x_1'' + 3x_1' + 2x_1 + x_2' + x_2 = 0, \\ x_1' + 2x_1 + x_2' - x_2 = 0, \end{cases} \qquad x_1(0) = 1,\ x_1'(0) = -1,\ x_2(0) = 0;$$

Formulas for Laplace Transform of Derivatives

$$\mathcal{L}\{y'(t)\} = sY(s) - y(0)$$

$$\mathcal{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0)$$

$$\mathcal{L}\{y'''(t)\} = s^3Y(s) - s^2y(0) - sy'(0) - y''(0)$$

$$\mathcal{L}\{y^{(4)}(t)\} = s^4Y(s) - s^3y(0) - s^2y'(0) - sy''(0) - y'''(0)$$

Table of Laplace Transforms

	$f(t)$	$\mathscr{L}(f)$
1	1	$1/s$
2	t	$1/s^2$
3	t^2	$2!/s^3$
4	t^n $(n = 0, 1, \cdots)$	$\dfrac{n!}{s^{n+1}}$
5	t^a (a positive)	$\dfrac{\Gamma(a+1)}{s^{a+1}}$
6	e^{at}	$\dfrac{1}{s-a}$

	$f(t)$	$\mathscr{L}(f)$
7	$\cos \omega t$	$\dfrac{s}{s^2+\omega^2}$
8	$\sin \omega t$	$\dfrac{\omega}{s^2+\omega^2}$
9	$\cosh at$	$\dfrac{s}{s^2-a^2}$
10	$\sinh at$	$\dfrac{a}{s^2-a^2}$
11	$e^{at}\cos \omega t$	$\dfrac{s-a}{(s-a)^2+\omega^2}$
12	$e^{at}\sin \omega t$	$\dfrac{\omega}{(s-a)^2+\omega^2}$